FAAV Build Manual

MIT AURA Lab, 2026 v4.0 (printable edition)

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1 Overview

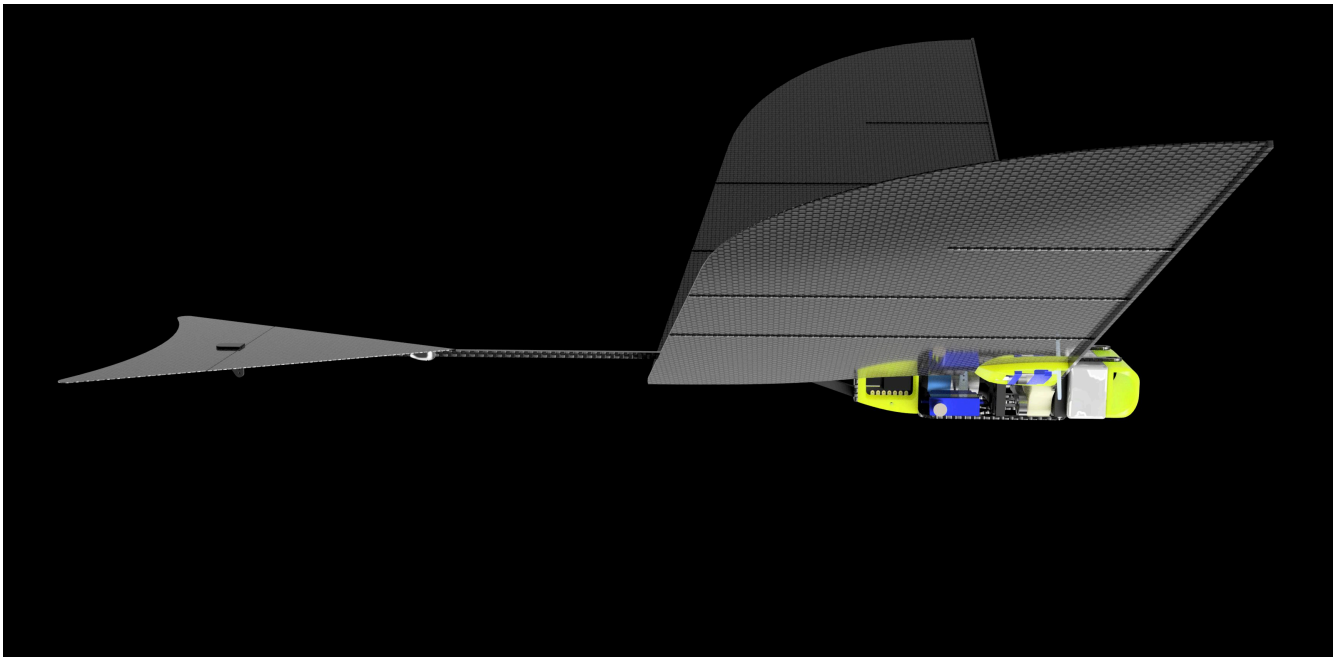


Figure 1: The FAAV3.2

Over the semester your team builds a **flapping aerial-aquatic vehicle**: a robot that flies, lands on water, swims by flapping the same wings, and takes off again. This manual is the build side of the course. It is sequenced to the weekly modules, and each page covers exactly what you do in that module.

1.1 How this manual is organised

Build modules are cumulative and in order. Each one opens with what you are building that day, what to have ready before you arrive, and ends with a checkpoint a TA signs off.

Reference holds the material you consult rather than follow: the parts list, the firmware repositories, the PCB revisions, the CNC guide, and the design notes explaining why the robot is the way it is.

1.2 Semester map

Module	Week	Build	Paired lecture
1	Sep 9	Orientation, teams, the robot	Intro to robotic locomotion
2	Sep 14	Motor and transmission shaft	Reaction forces in fluids
3	Sep 21	Motor driver, sensor, serial	Bird-scale flapping design
4	Sep 28	Closed-loop control and tuning	Swimming robots I

Module	Week	Build	Paired lecture
5	Oct 5	Frame, crank, linkage, gears	Undulatory locomotion
—	Oct 12	*No module*	—
6	Oct 19	Flapping simulation (Ptera)	Insect-scale flapping
7	Oct 26	Wing fabrication	Swimming robots II
8	Nov 2	Tail and housing	Multi-modal locomotion
9	Nov 9	Flight controller, PCB, dry testing	The future of actuation
10	Nov 16	Waterproofing	Multirotors
11	Nov 23	Wet testing	Fixed-wing flyers
12	Nov 30	Competition	Pair presentations

1.3 Conventions

> ⚠ Warning
>
> An operation whose likely failure mode destroys a part or injures you. Read it twice before acting.

> ℹ Checkpoint — TA sign-off
>
> Do not proceed past this without a TA looking at your work.

> 💡 Why
>
> Design rationale. Skippable at the bench, useful for your report.

1.4 Architecture in one paragraph

One 4S battery (16.8 V max) feeds a custom PCB that steps VBAT down to 5 V to power the flight controller, which in turn generates a 3v3 rail for the sensors on that PCB. Battery power passes through a shunt for current sensing, then out to the ESC. The flight controller commands the ESC over serial. The ESC drives a brushless motor through a 23.04:1 gearbox, and a crank–linkage converts that rotation into wing flapping.

1.5 Versioning

This is version **fall-2026**. Corrections published during the semester are dated in the Changelog. If a step here contradicts what a TA tells you at the bench, the TA is right and this page is stale — report it so the next team does not lose time to it.

2 Module 1 — Orientation

Week of Sep 9 · Paired lecture: *Intro to robotic locomotion* · Teams formed today

2.1 What you do today

Form teams, complete shop training, meet the robot you are going to build, and collect your parts kit.

2.2 The robot

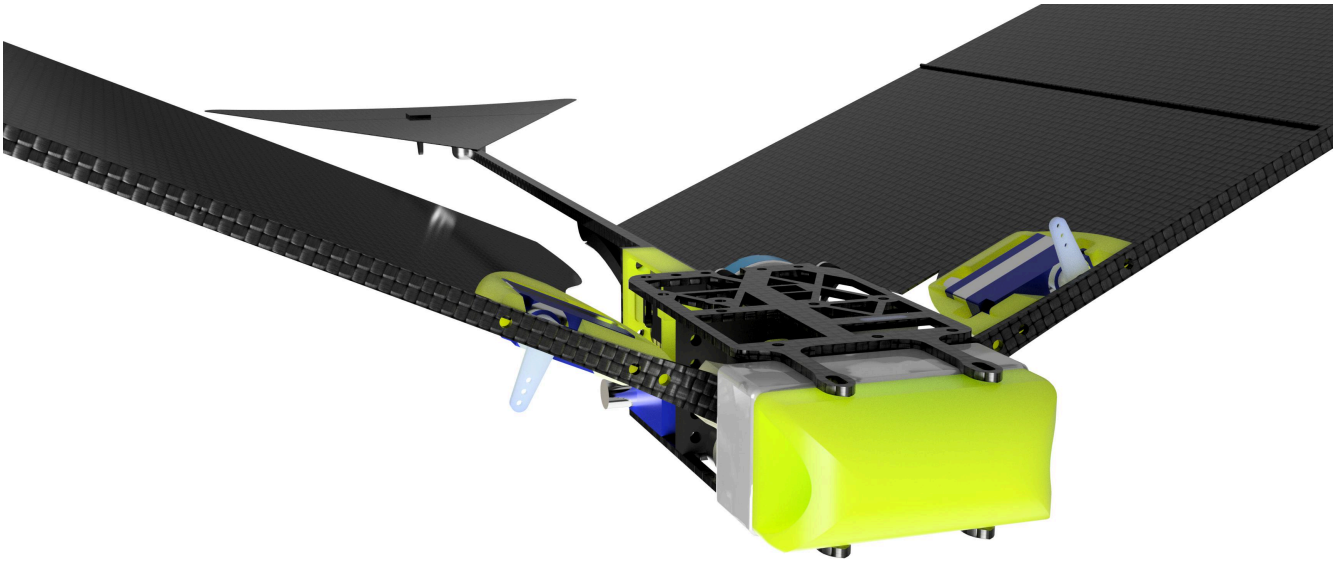


Figure 2: The FAAV3.2

The FAAV is a flapping-wing vehicle that operates in air and in water with the same actuation. The build divides into four subsystems, and the modules follow that order:

1. **Transmission** — motor, custom shaft, gearbox, crank and linkage (Modules 2, 5)
2. **Electronics** — ESC, angular sensor, flight controller, custom PCB (Modules 3, 4, 9)
3. **Aero/hydro surfaces** — wings, tail, foam housing (Modules 6–8)
4. **Waterproofing and testing** — potting, immersion, tuning (Modules 10–12)

2.3 Electronic architecture

One 4S battery (max 16.8 V) connects to a custom PCB that steps VBAT down to 5 V to power the flight controller. The flight controller generates a 3v3 rail for the sensors on the PCB. Battery power passes through a shunt for current sensing on the custom PCB, then back out to power the ESC. The flight controller sends commands to the ESC over serial.

2.4 Before you leave today

- ☐ Team formed and registered
- ☐ Shop training complete
- ☐ Parts kit collected against the parts list
- ☐ Access to the lab GitHub organisation confirmed

> ! To be completed by the instructor
>
> Shop safety requirements, machine access and booking procedure, team size and formation rules, and where the parts crib is. These are institution-specific and are not carried over from the manufacturing guide.

> i Checkpoint — TA sign-off
>
> Team registered, kit checked against the parts list, and every member has completed shop training.

3 Module 2 — Motor and transmission shaft

Week of Sep 14 · Paired lecture: *Reaction forces in fluids*

3.1 What you build today

A brushless motor with a custom shaft pressed into its rotor bell, paired to the Faulhaber gearbox, with the sun gear and the sensor magnet installed.

3.2 Before you arrive

Parts	AS2303 KV1500 motor, 4 mm steel rod, Faulhaber 23.04:1 gearbox, M2.5×10 screws
Tools	Circlip pliers, manual arbor press, lathe (booked), cutting oil
Read	This page, and the *Motor shaft* section below in full before touching the motor

3.3 How the transmission works

The motor is connected via a custom shaft to a gearbox with a **23.04:1** reduction by Faulhaber.[1] Both parts are held in place by 3D-printed plates. The output shaft of the gearbox has its own bearing on the inside of the gearbox, and is also supported mid-way on the shaft by a 4×8 mm flanged bearing. This leaves **5 mm of shaft length free** for the crank to be mounted on. A flat on the shaft helps lock the rotation of the M3 screw extending through the crank.

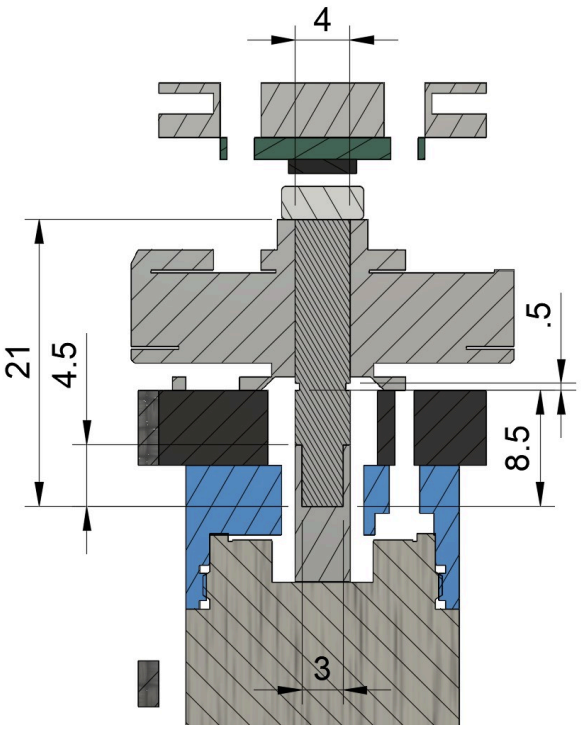


Figure 1: Motor shaft



Figure 2: Faulhaber gearbox

[1] Faulhaber gearhead reduction ratios (accessed 20.02.2024).

3.4 Which motor do you have?

3.4.a Black — AS2303 KV1500 (new)



The propeller horn **cannot** be separated from the bell. You will remove the two screws, then clamp the bell in a lathe using the stock shaft and machine it down (step 4 below).

3.4.b Blue — AT2303 KV1500 (old)



The horn is a separate part and comes off easily. Skip the lathe operation.

3.5 Motor shaft

To pair the motor with the gearbox, the shaft of the brushless motor needs to be replaced. This is a delicate operation — the motor's structure is mostly aluminium and quite thin, for weight savings.

1. Remove the circlip on the shaft with the special pliers. **Don't lose it, nor the brass ring.**
2. Separate the rotor from the stator delicately, by hand.
3. Press the shaft out of the rotor bell.

> ⚠ Never load the bell radially
>
> Press the shaft out very carefully, making sure to never apply forces radially on the bell. Use the manual press with appropriate support below the bell to avoid deforming the dome shape. **Do not squeeze the bell with a vice, or even by hand.**

4. Remove the propeller horn, which is not needed — see the tabs above for which case applies to you. On the black motor, machine the bell as follows:



Figure 3: Stock bell

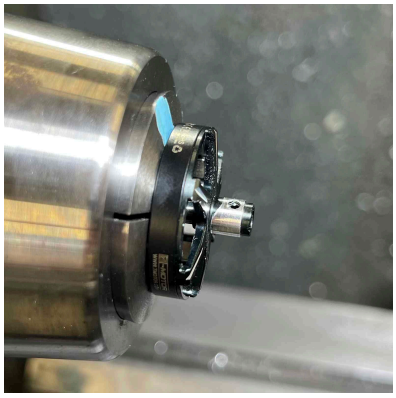


Figure 4: Bring diameter down to 6 mm

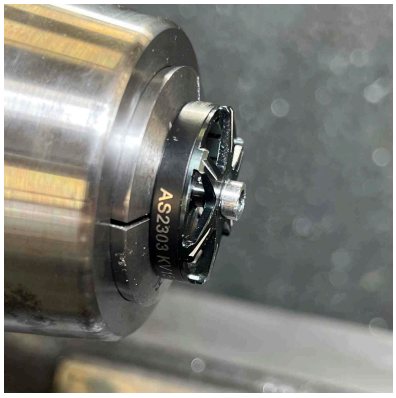


Figure 5: Bring the length down to 2.5 mm

5. Manufacture the new shaft from 4 mm diameter steel rod. It should have a **clearance fit** through the bearings and a **tight fit** on the rotor bell. See Figure 3 for exact dimensions.

> ⚠ Check your wall thickness before cutting
>
> 21.5 mm is the shaft length **if the motor and the gearbox are separated by a 5.5 mm-thick wall**. If your wall is 4 mm thick, reduce the 8.5 mm segment to 7 mm, bringing the total length from 21.5 mm to 20 mm.

6. Press the new shaft into the bell, making sure not to press on the dome-shaped part of the bell without appropriate support underneath.

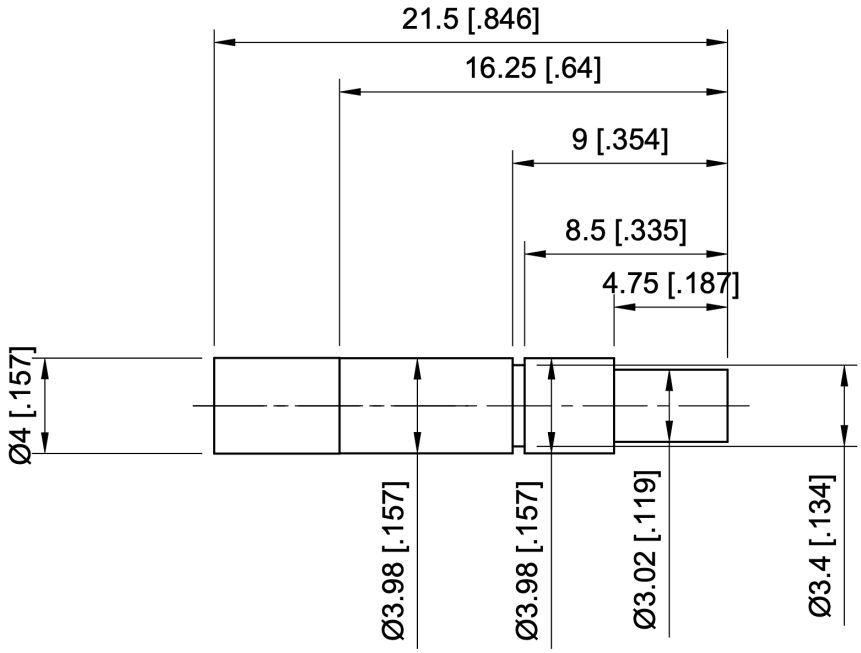


Figure 3: Motor shaft dimensions

3.6 Gearbox assembly

The blue part of the gearbox is fixed on one side using the custom screws that come with the gearbox, and with **M2.5×10** screws on the other.

> ! To be completed by the instructor
>
> Sun gear installation and sensor magnet bonding are in the course project plan but are not documented in the manufacturing guide. Add the magnet part number, bonding agent, and the concentricity tolerance that matters for Module 4.

> i Checkpoint — TA sign-off
>
> Shaft seated, rotor spinning freely by hand with no wobble, gearbox mounted, and the sensor magnet concentric with the shaft axis. An off-centre magnet produces sensor faults that are very hard to diagnose two modules later.

4 Module 3 — Motor driver, sensor and serial

Week of Sep 21 · Paired lecture: *Bird-scale flapping design*

4.1 What you build today

A flashed G431 ESC talking to your laptop over serial, with the angular sensor wired on SPI3 and the motor spinning under commander control.

4.2 Before you arrive

Parts	B-G431B-ESC1 board, AS5047P sensor, ST-Link programmer, wire

Tools	Soldering iron, fine tweezers, multimeter
Read	This page, plus Design notes on why we use an FOC ESC

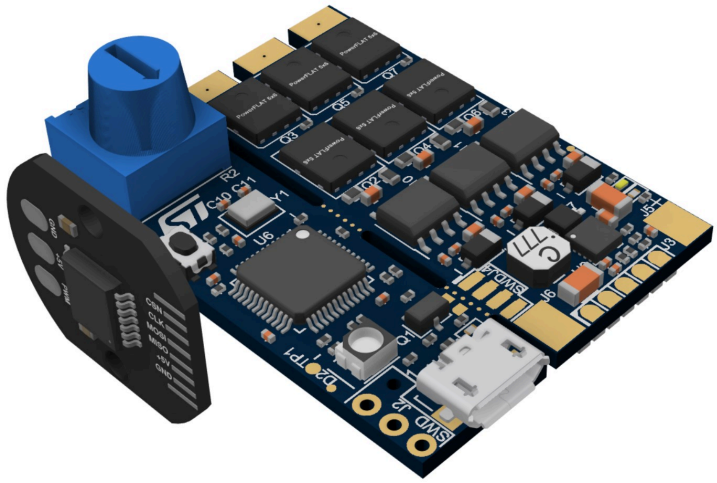


Figure 4: ESC running SimpleFOC, with the ST-Link programmer and the angular sensor

4.3 Firmware

Code for the B-G431B-ESC1 motor controller, using the SimpleFOC library, is in the lab repository: FAAV-G431-ESC. See Software for all repositories.

4.4 Commanding and monitoring

This is a high-level summary of the SimpleFOC communication docs. Communication with the ESC is over **serial**, either via the ST-Link programmer (the small attached board) or via a separate microcontroller board.

Once serial is established at **115200 baud**, you interact with the ESC through the **commander** — a practical interface for setting and getting values (instructions). Ask the commander for available commands with `?`. Currently available:

Command	Effect
`M`	Full motor configuration
`F`	Get the main loop frequency
`L`	Turn the onboard LED on and off (useful as a comms check)
`T`	Set the target value — rad, rad/s or N/m depending on control mode
`W`	Watchdog on/off. If no new target arrives for 1 s, target is set to zero. **On by default**
`R`	`R0`/`R1` disable/enable a linear ramp when changing set-point. `RD1000` sets ramp duration to 1000 ms (default 1000 ms). Off by default

> ⚠ Careful when changing control mode
>
> `T` means something different in each mode — position, velocity, or torque. Sending a velocity setpoint while in torque mode will do something you did not intend, with the wings attached.

4.5 ST-Link serial

The ST-Link needs to be connected to the ESC MCU. Out of the box that happens via **Serial2**, but not once those GPIOs are used as SPI3 for the sensor. In that case you must connect **Serial1** to the ST-Link:

1. Disconnect USART2 and connect USART1: remove **R23** and **R24** on the daughter board (desolder).
2. Solder wires to the now-free pins of R23/R24 — the pads closer to U3, further away from the USB port. Connect the wire from **R23** to the pad for hall sensor 2 (**PB7**), and the wire from **R24** to the pad for hall sensor 1 (**PB6**) on J8.
3. Remap the pins in software for SPI use: in `platformio.ini` via `build_flags`, and by overwriting the `WEAK PinMaps` in `pinmap_custom.h`. See commit `18520ac` on GitHub for details.

4.6 Angular sensor on SPI3

There is no SPI link exposed directly on the board. The solution is to use the **SPI3 bus**, exposed through a few hardware modifications and matching software changes.

Table 1: SPI3 connections on the G431. * Solder onto the outward side of the resistor.

	SCLK	MISO	MOSI	CS
MCU GPIO	PB3	PB4	PB5	PA15
G431B pin	J3	J3	R14*	J3

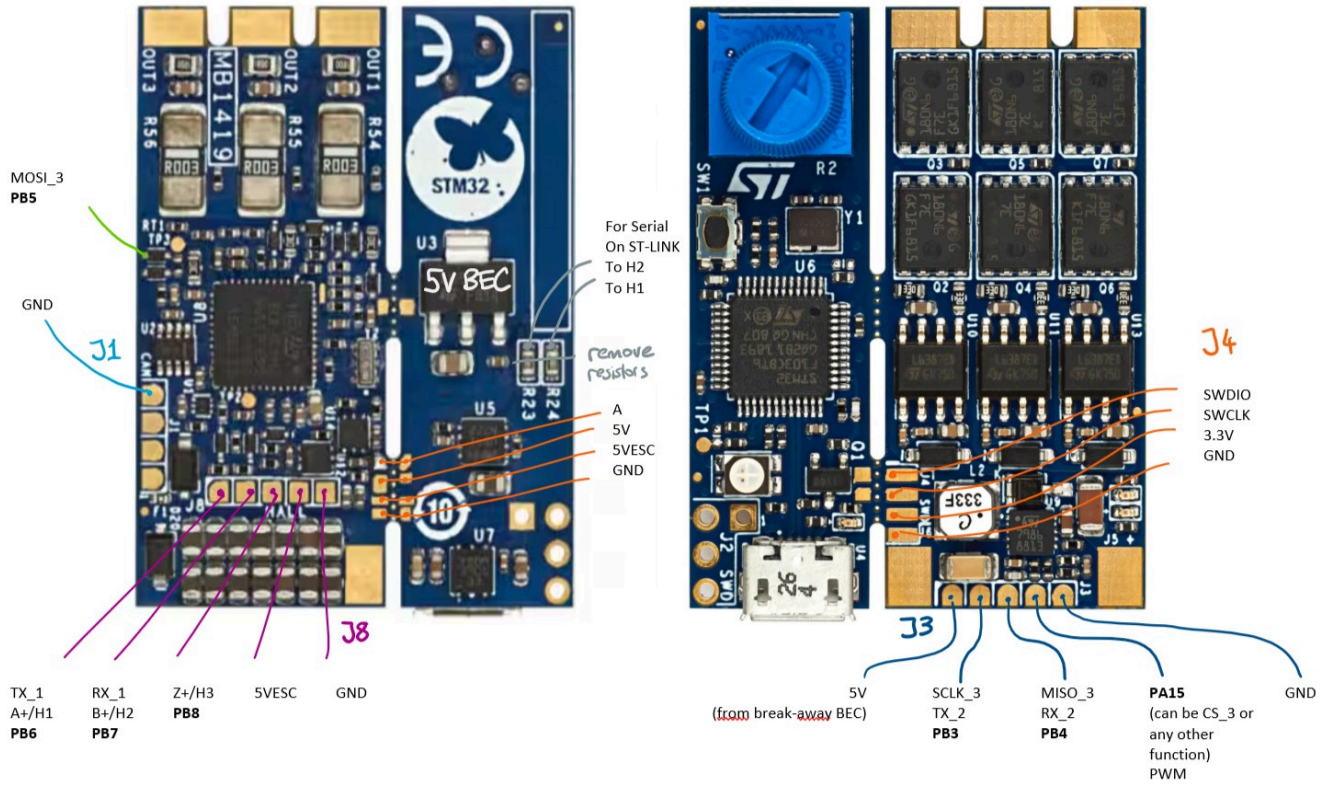


Figure 5: GPIOs of the G431B-ESC1 board

ℹ Checkpoint — TA sign-off

Commander responds to `?` over serial at 115200 baud, `L` toggles the onboard LED, the sensor returns a plausible angle through a full manual rotation, and the motor spins under a `T` command.

5 Module 4 — Closed-loop control and tuning

Week of Sep 28 · Paired lecture: *Swimming robots I*

5.1 What you build today

The gearbox installed with a marker flag on the output, running continuously, then holding position, then tracking a velocity setpoint with a tuned loop.

5.2 Before you arrive

Parts	Assembled motor + gearbox from Module 2, ESC from Module 3
Tools	Bench supply with current limit, tachometer or camera
Read	This page in full — the speed limits below explain most of what you will hit

5.3 Task sequence

1. Mount a small flag on the gearbox output so rotation is visible and countable.
2. Achieve continuous spinning at low speed.
3. Achieve position control — the output holds against a hand disturbance.
4. Tune the loop and characterise your maximum usable speed.

! To be completed by the instructor

The specific PID tuning exercise, the target performance, and the sign-off criteria are course material and are not in the manufacturing guide.

5.4 Why low speed is the hard part

Regular drone ESCs start **losing synchronisation at low speed**, which is exactly the regime used for water propulsion. That is the reason this course uses an FOC ESC rather than a hobby unit — see Design notes.

5.5 Going beyond 600 rad/s

Two issues limit speed: the micro-controller loop rate, and delays from the rotation sensor, which also feed back into the first. The changes below improved maximum speed from **500 to 1700 rad/s**, enough to drive the wings at 10 Hz. At very high speeds sensor lag becomes the binding constraint, and shows up as high current draw and noise.

5.5.a Sensor speed

- Switch to an **SPI sensor**. SPI is faster than I²C or PWM. There is no SPI link directly exposed on the board — the solution is the SPI3 bus, wired as in Module 3.
- Increase SPI bus speed to **10 MHz**.
- Use the **AS5047P** sensor, rated for speeds up to 28 kRPM.

5.5.b MCU speed

The goal is to maximise main loop frequency, which reached **13 kHz**. Read the current loop frequency with the commander's `F` command.

Change	Effect
`motor.motion_downsample = 3`	+0.8 kHz loop frequency
Disable monitoring	+0.2 kHz
Switch to SVPWM	No improvement observed. May help, unconfirmed
`L = 0.0001`	Tuning the inductance value helped — made voltage mode work much better

5.5.c Sensor lag

Compensating sensor lag as speed increases is necessary to go above 500–600 rad/s without wasting current. It is currently hand-tuned every 100 rad/s, and a fitted function is used on the robot. This could be automated by ramping through the speeds and minimising phase current — future work. This SimpleFOC forum thread is useful on the topic.

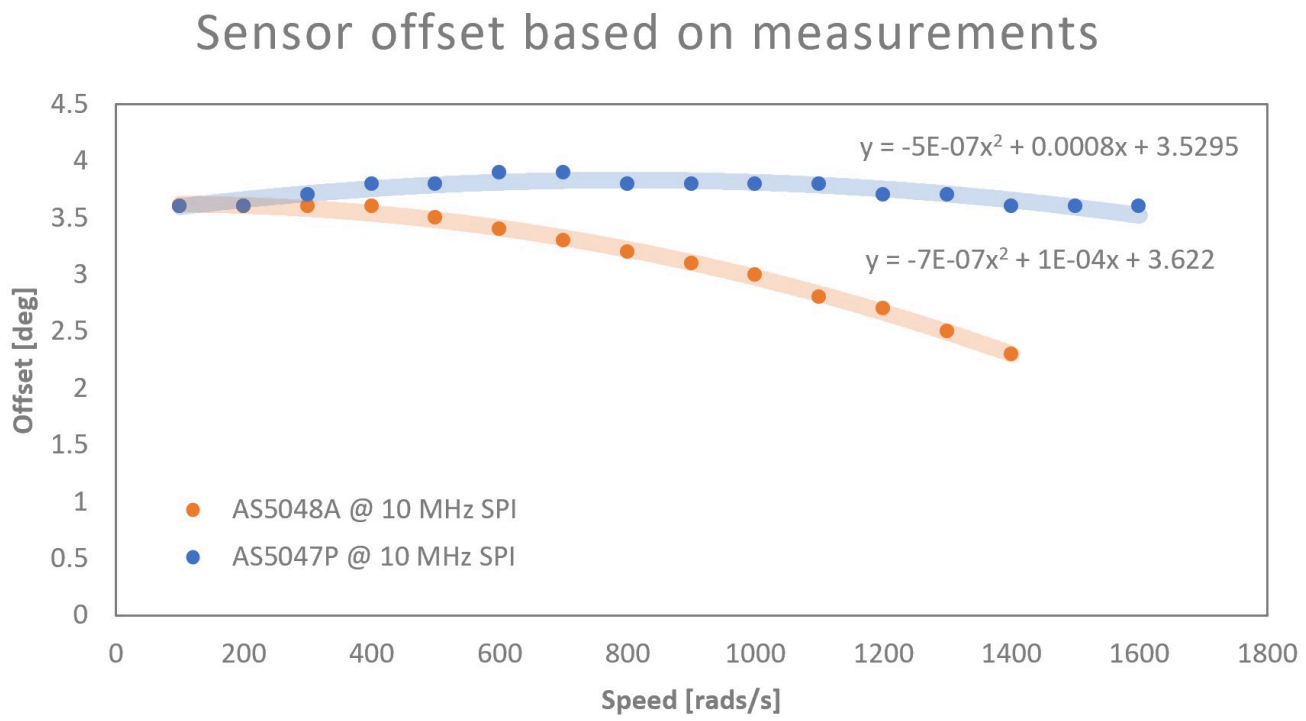


Figure 6: Optimal sensor offset against speed, for two sensors

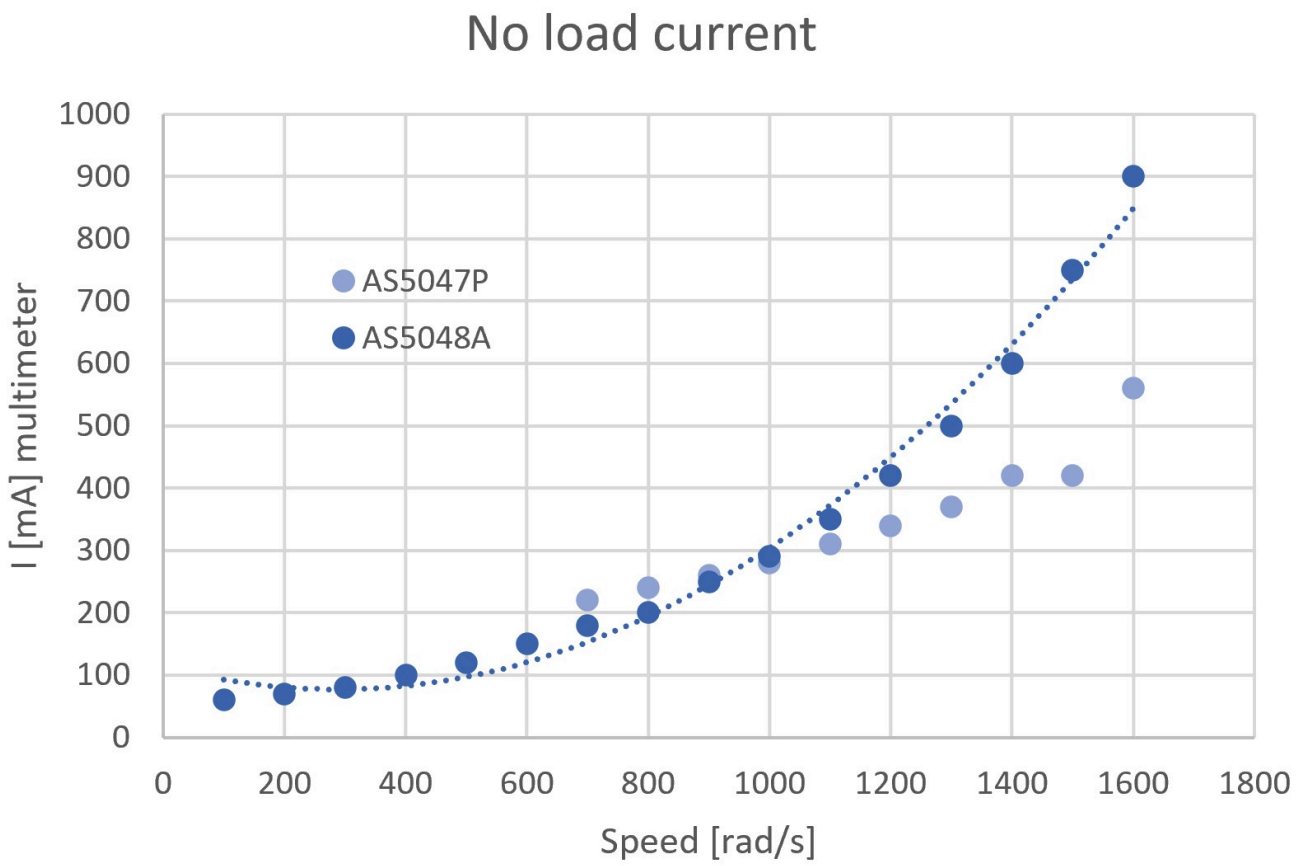


Figure 7: Current draw with two different sensors, after compensation

Averaging current measurements

When using the ESC's current measurements (I_{DC}, I_d, I_q, I_u, I_v, I_w), always average over a time window — the raw measurements are noisy.

Checkpoint — TA sign-off

Continuous low-speed rotation without loss of synchronisation, position hold against a hand disturbance, and a recorded loop frequency from F.

6 Module 5 — Frame, crank, linkage and gears

Week of Oct 5 · Paired lecture: *Modelling of undulatory locomotion*

6.1 What you build today

The carbon frame, the CNC-cut crank and linkage, the PEEK gears, and the full mechanical integration that turns motor rotation into flapping. Flapping amplitude is fixed by this design.

6.2 Before you arrive

Parts	1.5 mm woven carbon plate, 5 mm aluminium plate, 5 mm PEEK plate, bearings, M2×4 screws, brass inserts

Tools	CNC (booked), drill press, M3 tap, cutting oil, metal glue
Read	This page, plus the CNC guide if you have not run the Stepcraft before

6.3 Main body

The main body is a carbon sandwich, spaced **24 mm** apart. The plates are CNC-cut from woven 1.5 mm plates, available from swiss-composites; thickness usually varies from 1.5 to 1.6 mm. The top and bottom plates are identical, except for the chamfers needed for the M2 screws. The screws holding plate to plate are **M2×4**, screwing into brass inserts.

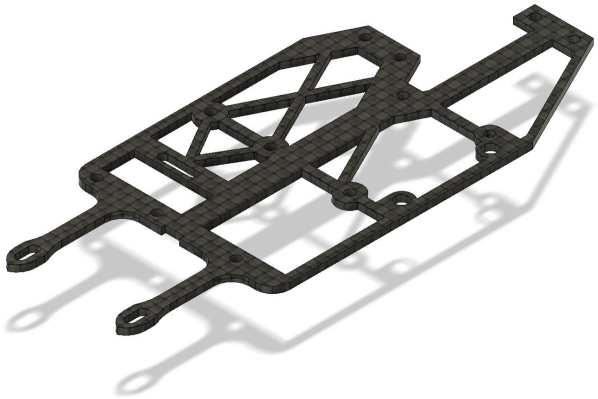


Figure 8: Main carbon frame

Fusion360 generates cutpaths from the parts in the *manufacture* tab and allows for *manufacturing models*, where parts can be recombined, realigned and even modified. Here, chamfers were added in the manufacturing model. Operations go: **holes first, then chamfers, then pockets, and finally the outline.**

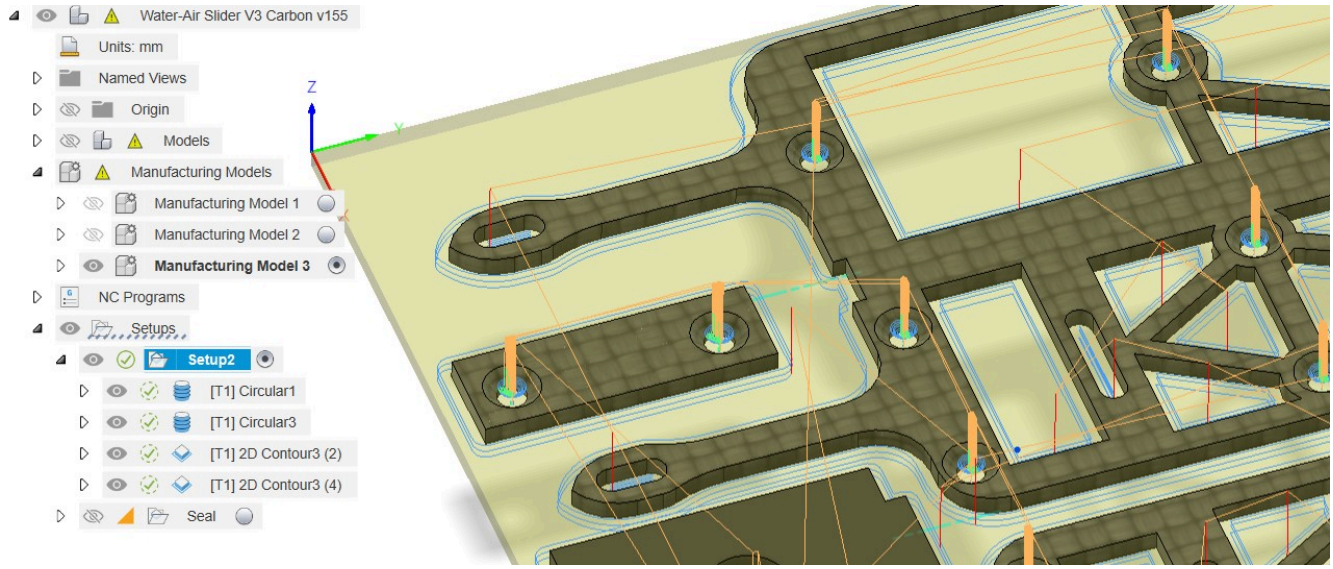


Figure 9: Manufacturing model

> ⚠ Check your zero before you cut
>
> Make sure the machine's 0 is at the right location in your file, and check the orientation of the axes. Full CNC guide for the Stepcraft D660 is in the CNC reference.

6.4 Crank and linkage

Dimensions differ between robot generations — check which one you are building:

Table 2: Crank–linkage dimensions

	FAAV3.2	FAAV4
Date	2023	July 2024
Crank radius	4 mm	4.5 mm
Linkage length	23 mm	24 mm
Crank–linkage shaft diameter	3 mm	4 mm
Bearing, crank side	10×3×4	10×4×4
Bearing, wing side	8×3×4	8×3×4 flanged

Both the crank and linkage are CNC-cut in a 5 mm aluminium plate. The linkage is only 4 mm thick, cut down from 5 mm with a facing operation on the CNC. Use a **2 mm endmill**. Flooding the part keeps the temperature down and gives good results.

Once cut:

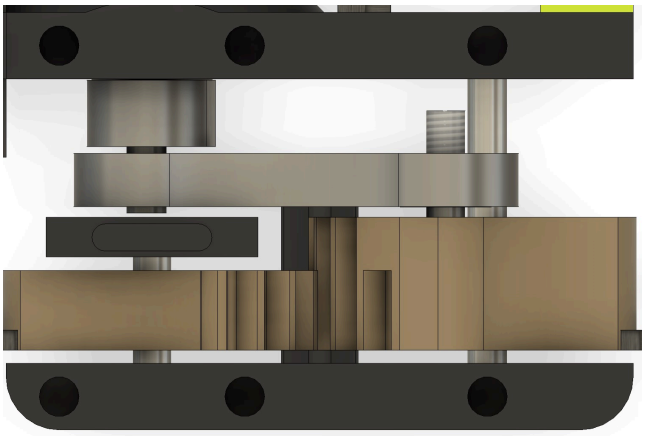
1. Manually drill a **2.5 mm** bore hole into the crank, perpendicular to its 4 mm hole and to the flat raised surface. Drill through the part all the way to the 4 mm hole.
2. Carefully tap an **M3** thread. Use cutting oil and make sure to break off the chips.

> ⚠ This thread carries high forces
>
> It **must** be done carefully, keeping the alignment between hole and tap.

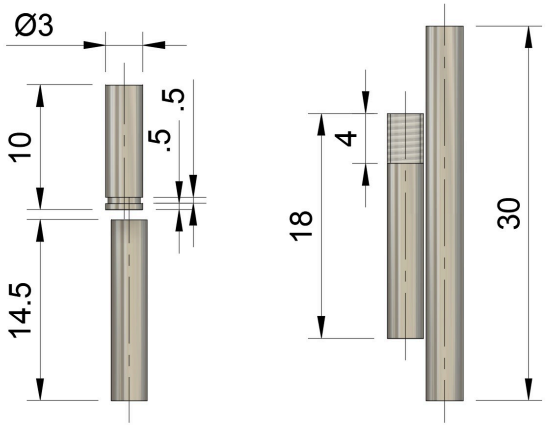
3. Press the bearings into the linkage. They may need to be bonded with metal glue if the press fit is not good enough.
4. Join the crank and linkage with a short 3 or 4 mm shaft, cut to **9.5 mm**. The bearing is kept from coming out with a pointy indent on the shaft end. *(To be improved.)*

> ⚠ Alignment of the linkage, perpendicular to the shafts, is extremely important
>
> An offset of **1 mm** is sufficient to destroy the crank-link in a few flights. To ensure forces are not applied radially, use circular offsets (not shown in the schematics) that fit around the shafts and support proper alignment of the linkage and shafts.

Figure 10 shows how the four shafts for the FAAV3.2 should be manufactured. Nothing exotic, but they need to be accurate.



(a) Transmission, top view



(b) Shaft dimensions

Figure 10: Shafts

6.5 Gears

Cut in **5 mm PEEK** plastic — a high-performance plastic suited to hard environments and high temperatures. The 5 mm plate comes from RS Components. Cutting underwater again gives good results. Keep the advance speed low enough to cut the teeth cleanly.

The primary gear consists of **2 parts**, stacked and screwed together. The 3 mm holes where the carbon is fixed need to be **tapped** with a sharp M3 tap.

> Keep the tap normal to the surface
>
> Best done in the drill press, which keeps the tap aligned.

The 3 mm shaft is pressed through the two gears' centre holes — a mildly tight fit at minimum, otherwise glue the shaft. Same for the linkage shaft (also 3 mm), which should be bonded as well as possible; this one matters more, so use glue in all cases.

Between the gears, a clearance of **0.4 mm** (inter-axis spacing of 25.4 mm) ensures proper operation without too much backlash.

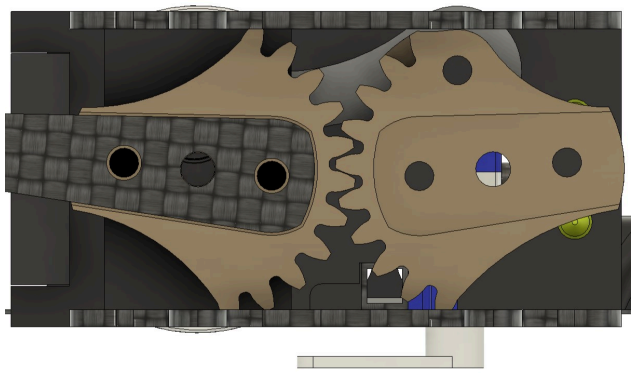


Figure 6: Transmission assembled

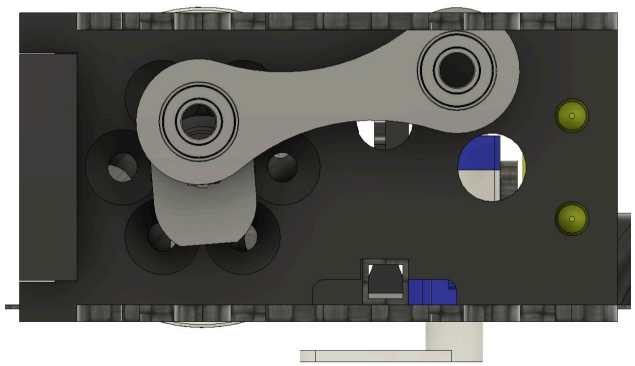


Figure 7: Transmission, second view

ℹ Checkpoint — TA sign-off

Frame assembled, crank–linkage rotating freely through a full cycle with no binding, linkage visibly perpendicular to the shafts, and gear backlash checked.

7 Module 6 — Flapping simulation (Ptera)

Week of Oct 19 · Paired lecture: *Insect-scale flapping* · No module Oct 12

7.1 What you do today

Choose your wing and tail geometry. You simulate candidate geometries in Ptera and pick the one that gives the best thrust profile when compounded with the provided motor power curve. Modules 7 and 8 build whatever you decide here.

7.2 Before you arrive

Bring	Laptop with Ptera installed, your measured loop performance from Module 4
Read	This page

7.3 Deliverable

A wing planform and a tail geometry, with the simulation output that justifies them, carried forward into Module 7 and Module 8.

! To be written by the instructor

This module has **no counterpart in the manufacturing guide** — the guide documents the research robot's final geometry, not the design exercise. The following need to be written from course material:

- Ptera installation and a worked example
- The provided motor power curve, and its format
- What "most efficient thrust profile" means quantitatively here, and how it is scored
- Constraints the students must respect so their geometry remains buildable with the frame from Module 5 — span limits, root fitting, mass budget
- Whether teams may deviate from the reference wing, and by how much

ℹ Checkpoint — TA sign-off

A geometry chosen, with simulation evidence, and confirmed buildable against the frame and wing-root hardware.

8 Module 7 — Wing fabrication

Week of Oct 26 · Paired lecture: *Swimming robots II*

8.1 What you build today

Both wings: carbon skeleton, membrane cut to the geometry you chose in Module 6, bonded and stitched at the load-bearing points.

8.2 Before you arrive

Parts	Carbon rods and strips, wing-root fittings, membrane material, glue
Tools	Cutting mat, steel rule, scalpel, printed full-size template of your geometry
Bring	Your Module 6 geometry, printed 1:1

8.3 Wing roots

The wing roots are carbon, drilled to mate with the linkage bearings from Module 5.

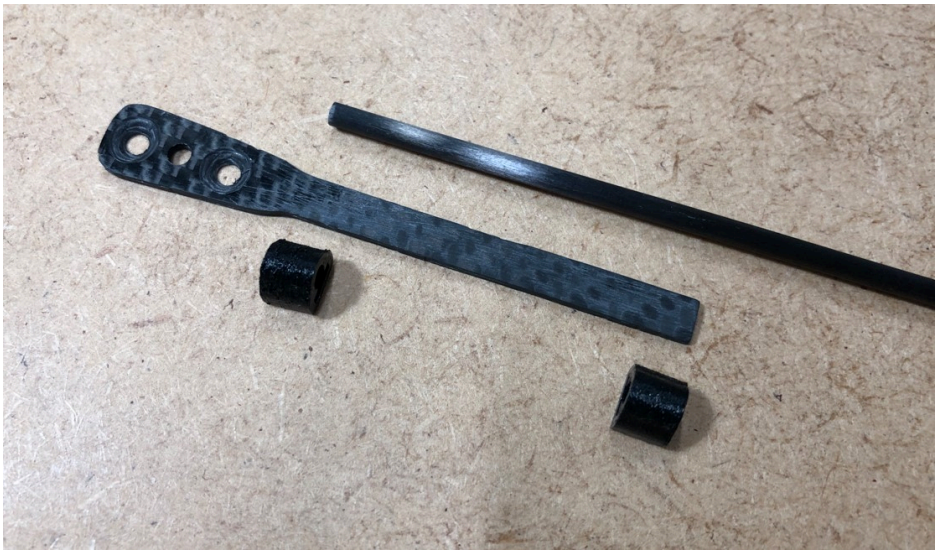


Figure 11: Wing root parts and spar stock

8.4 Laying out and cutting the membrane

Work over the full-size printed template, taping the spars in position before cutting the membrane to the outline.

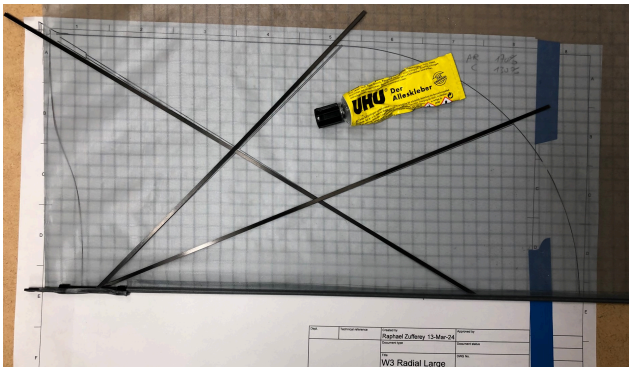


Figure 8: Spars taped over the template

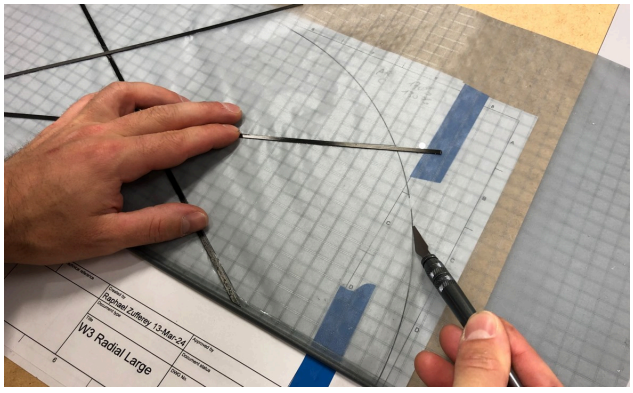


Figure 9: Cutting the membrane to the outline

8.5 Completed wings

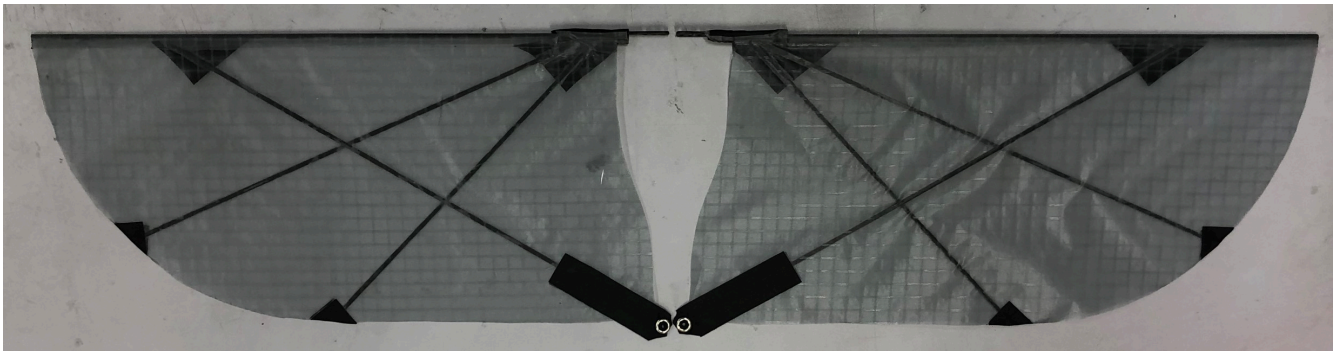


Figure 12: Completed wing pair

> ⚠ Both wings must match
>
> Asymmetry between left and right shows up as a roll bias you will spend Module 11 trimming out. Weigh both wings and compare spar positions against the template before bonding.

> ! To be written by the instructor
>
> The manufacturing guide documents this step **with photographs only** — there is no written procedure in the source. The images above are in build order, but the following still need writing:
>
> - Membrane material and its supplier
> - Adhesive, and cure time
> - Which areas are stitched, and with what
> - Spar layup: rod diameters, where the diagonals attach
> - Tolerance between left and right wing

> i Checkpoint — TA sign-off
>
> Both wings complete, matched in mass and geometry, roots fitting the linkage bearings without play.

9 Module 8 — Tail and housing

Week of Nov 2 · Paired lecture: *Multi-modal locomotion*

9.1 What you build today

The passive tail, cut to the shape your simulation produced and mounted on its rod, plus the foam housing that protects the robot and keeps it afloat.

9.2 Before you arrive

Parts	Carbon sheet and tail rod, foam stock (XPS or closed-cell PE), epoxy
Tools	Foam cutter, scalpel
Bring	Your Module 6 tail geometry, printed 1:1

9.3 Tail

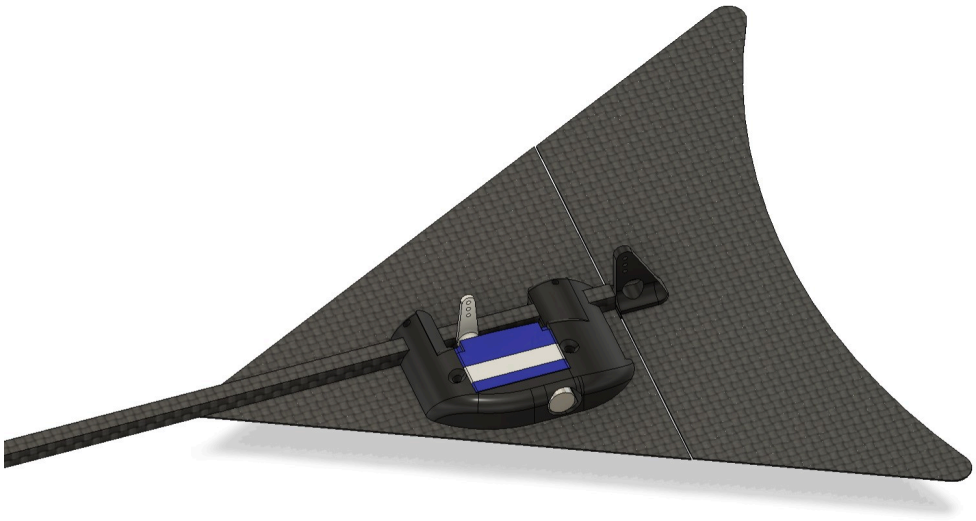


Figure 13: Tail assembly

> ! To be written by the instructor
>
> As with the wings, the guide contains **only the photograph above** for the tail. Still needed: sheet material and thickness, rod diameter, how the tail mounts to the boom, and the incidence setting. The project plan also notes tail actuation is optional if time permits — decide before the semester whether that is in scope.

9.4 Housing

Use a foam cutter to cut an exterior housing for the FAAV. This is an important step for external tests: it adds a layer of protection and ensures the FAAV floats. Leave space for GPS and antenna insertions if you are using them.

To keep the FAAV level when floating — which helps visibility and getting it out of the water — it is recommended to put a certain amount of foam on the tail.

9.4.a Choosing a foam

Various foams have been tested, with tradeoffs:

Foam	Floatability	Absorption	Impact behaviour
Extruded polystyrene (XPS)	Best, low density	Zero	Relatively brittle; little help damping landings
Closed-cell polyethylene (PE)	Decent, relatively low density	Light	Much more compliant; usable as protection

If someone ever has time, a multi-material housing could have its use cases.

> ⚠ Check solvent compatibility before painting or gluing foam
>
> Some foams are very sensitive to solvents. Epoxy is inert and can act as glue.

ℹ Checkpoint — TA sign-off

Tail mounted and aligned, housing fitted with clearance for the wings through the full flap cycle, and the assembled robot floating level in the test tank.

10 Module 9 — Flight controller, PCB and dry testing

Week of Nov 9 · Paired lecture: *The future of actuation*

10.1 What you build today

The custom PCB assembled and validated, the flight controller flashed and talking to the ESC, and the robot flapping in air under wireless control.

10.2 Before you arrive

Parts	FAAV PCB v3.0, stencil, solder paste, XIAO ESP32S3, JST connectors, battery
Tools	Hotplate, tweezers, multimeter, bench supply with current limit
Read	This page and PCB revisions

10.3 Electronic architecture

One 4S (max 16.8 V) battery is connected to a custom PCB that steps VBAT down to 5 V to power the flight controller. The flight controller generates a 3v3 rail to power the sensors on the custom PCB. Battery power goes through a shunt for sensing on the custom PCB, then back out to power the ESC. The flight controller sends commands to the ESC over serial.

10.4 Flight controller



Figure 14: Microcontroller in its waterproofing box

Firmware lives in FAAV-Flight-Controller (ESP32, FreeRTOS tasks). See Software for older variants.

10.5 Assembling the PCB

As of April 2026 the current board is **v3.0**: 4 layers, 21 × 28.5 mm, 34 components. The complete BOM is in the Manufacturing/Assembly folder of the FAAV-PCB-v3 repository.

The simplest way to assemble a FAAV PCB is with a stencil, solder paste applied with a cutter blade, then reflow on the hotplate.

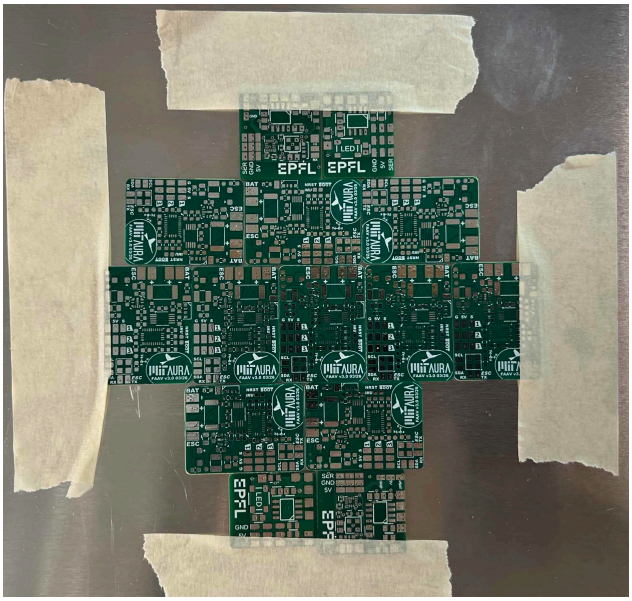


Figure 10: Tape blank PCBs around the PCB to assemble

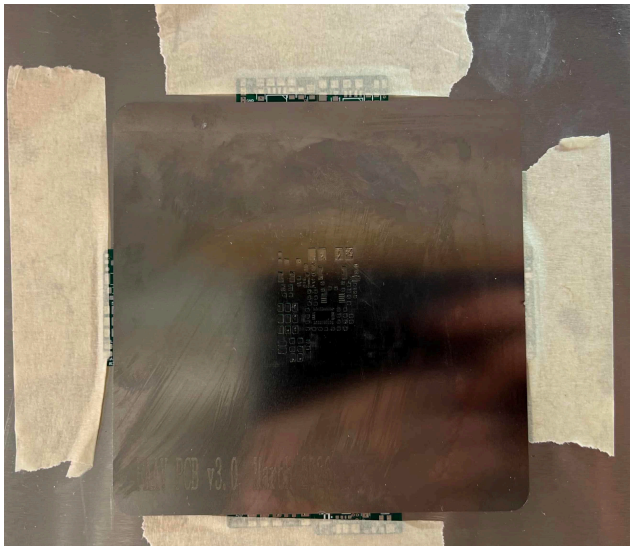


Figure 11: Align the stencil on top and tape it, making sure it sits flat

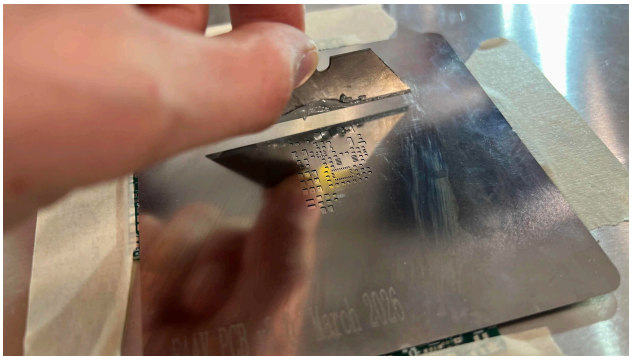


Figure 12: Wipe down solder paste with a blade in one firm pass



Figure 13: Place the assembled PCB on the hot plate

10.6 Taking the PCB into operation

Once assembled, solder power wires for VBAT and GND at the input side of the board. Follow these steps **before powering the PCB for the first time**:

1. Check that there is no short circuit between GND and 5V / 3.3V / VBAT.
2. Connect a **15 V supply with a 100 mA current limit** to the battery pads and check that it does not draw excessive current.

3. Check with a multimeter that the buck regulator outputs 5 V from the 15 V input. The MCU should be on by then.

With the MCU powered, flash your test firmware and check the connection to all sensors. Table 3 gives the I²C addresses.

Table 3: I²C addresses

Component	I²C address (bin)	I²C address (hex)
Pressure sensor	`0b1110110`	`0x76`
IMU	`0b1101000`	`0x68`
INA228 power monitor	`0b0101000`	`0x40`

Verify:

- ☐ Voltage, current and temperature readings from the INA228
- ☐ IMU readings
- ☐ Pressure sensor readings
- ☐ RGB LED control

Once all the hardware on the board is validated, solder the power wires at the output side of the board to the ESC, and the UART wires.

> ⚠ TX to RX
>
> The **TX** output of the PCB has to be connected to the **RX** pin of the ESC, and vice versa.

10.7 Connectors

The waterproof connector is the JST 8-pin 08R-JWPF-VSLE-D / 08T-JWPF-VSLE-D. All pins marked `_b` are used for balancing the battery during recharge. For use on the robot, connect at least **Vbat** and **GND**; to allow higher currents, connect **Vbat_b** and **GND_b** as well.

> ⚠ Seal the unused cable slots
>
> Do not forget to seal the back of the connector for the unused cable slots — this is a water ingress path straight into the electronics.



Figure 15: Connector wiring scheme. Left: battery connector. Right: connector on the robot / for recharging the battery.

10.8 Dry testing

> ! To be written by the instructor
>
> The dry-test protocol is course material, not in the guide. Needed: what the robot is mounted on while flapping in air, the flap frequency sweep, what is measured, the wireless control setup students reprogram to, and abort criteria.

> ℹ Checkpoint — TA sign-off
>
> PCB validated against all four sensor checks, flight controller commanding the ESC over serial, and a supervised flapping run in air.

11 Module 10 — Waterproofing

Week of Nov 16 · Paired lecture: *Multirotors*

11.1 What you build today

A potted PCB, a lubricated and sealed gearbox, and a robot that survives a static immersion test.

> ⚠ Silicone cure time gates this module
>
> The PCB potting takes **12–24 hours** to cure because of the depth. Plan the module so potting happens first and soldering happens after the cure, or across two sittings.

11.2 Before you arrive

Parts	Waterproofing box, Kapton tape, silicone sealant, USB-C cable, CorrosionX, white lithium grease
Tools	Needle, soldering iron
Read	This whole page before starting — the ordering matters

11.3 PCB waterproofing

First ensure that the assembled PCB fits within the box before anything else.

1. Tape the bottom side of the box with Kapton.
2. Fill the bottom of the box with silicone sealant until it reaches the USB port hole.
3. *Carefully* slide the PCB assembly in, with the Seeeduino facing the bottom of the box.
4. **Very important!** Insert a USB-C cable into the port. This keeps the sealant from flowing into the port, and the cable is a convenient point to hold the assembly by.
5. Add sealant along the three free edges to fill any gaps, until the sealant reaches the level of the top side of the PCB. **The PCB components and contacts should be left exposed.**
6. Let the silicone cure — because of the depth this takes about 12–24 h.
7. Solder the wires and contacts.
8. Solder the voltage regulator.
9. Seal the top side of the PCB. The best method here is unclear. Tested solutions:
   - Nail polish
   - Silicone sealant
   - 422C silicone conformal coating (datasheet)

11.3.a If you are on a v2.0 board

The method changes slightly. See PCB revisions for the v2.0 procedure, including the SD-card trap-door recovery and antenna potting.

11.4 Gearbox

By default the gearbox on the FAAV4 robot is **not waterproof at all**, and it is normal for it to end up completely filled with water once submerged. Over years of use the gearbox proved largely immune to corrosion — after more than a year of repeated use in water, including in Lake Geneva, one was opened and showed no sign of corrosion.

That is not the whole story, and the exception is the reason for this step:



Figure 16: Gearbox and sun gear after a few dives with motor oil lubrication

Severe corrosion appeared on a newer robot after only 3–4 uses in water. The cause is believed to be improper lubrication after cleaning, combined with the absence of the shaft bearing, which let more water in through the front. Motor oil had been used as a substitute lubricant; its low viscosity is believed to have washed it away quickly, leaving the gears dry. Full investigation in Design notes.

> ⚠ Use the right lubricant
>
> A mix of **CorrosionX and white lithium grease**. Do not substitute motor oil. Several dives have since been performed with the same gearbox that previously corroded, and it now runs smoothly.

Parts that rust: **sun gear, planet gears, ring gear, planet shafts.** Parts that do not: screws, motor shaft, washers, body, circlip.

11.4.a Sealing modifications

A few optional modifications to the FAAV structure and assembly process keep most of the water out.

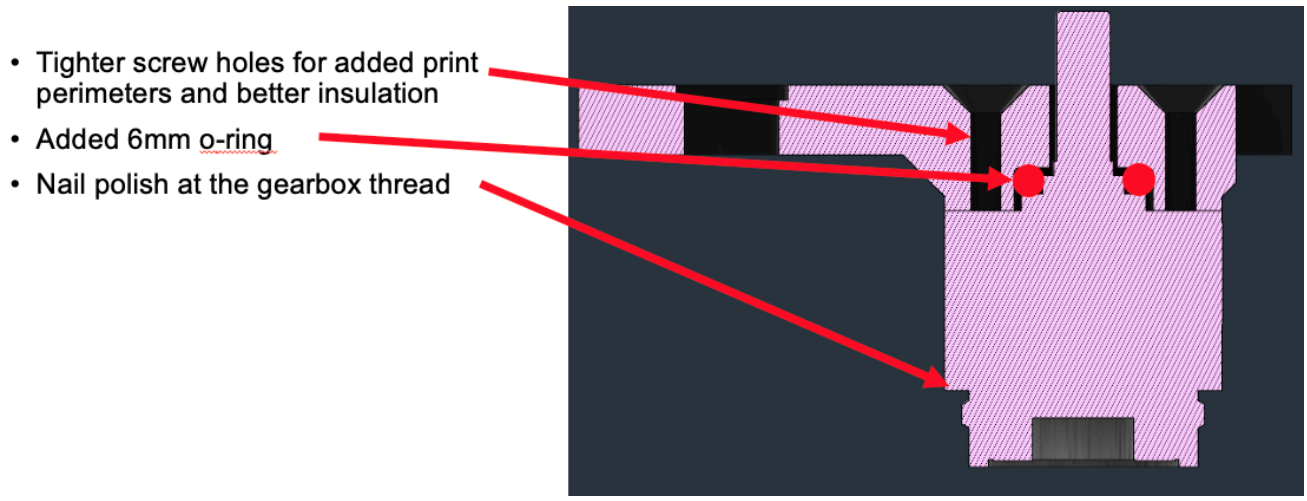


Figure 14: Gearbox waterproofing, part 1

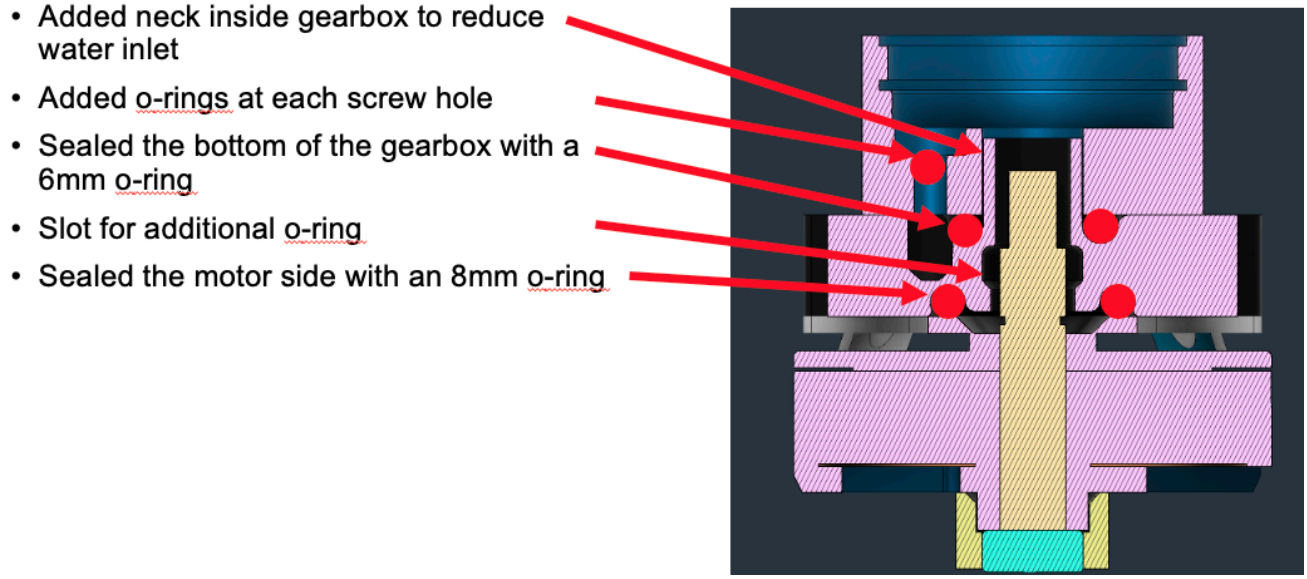


Figure 15: Gearbox waterproofing, part 2

> **i** Checkpoint — TA sign-off
>
> PCB fully cured and potted with contacts exposed, gearbox lubricated and reassembled with the shaft bearing fitted, connector back-shell sealed, and a static immersion test passed with no ingress.

12 Module 11 — Wet testing

Week of Nov 23 · Paired lecture: *Fixed-wing flyers*

12.1 What you do today

Immersion, underwater runs, wing trim, and control tuning in water — the regime the whole transmission was designed around.

12.2 Before you arrive

Bring	Fully waterproofed robot from Module 10, charged 4S pack, spare crank–linkage shaft
Read	This page, and your notes from Module 4

12.3 Pre-immersion checks

- ☐ Static immersion test from Module 10 passed, robot opened and inspected dry
- ☐ Connector back-shell sealed on all unused slots
- ☐ Gearbox lubricated with CorrosionX and white lithium grease
- ☐ Watchdog enabled on the ESC (`W`) — target drops to zero if commands stop
- ☐ Robot floats level

> ⚠ Low speed is where hobby ESCs fail
>
> Water propulsion runs the motor in exactly the regime where synchronisation is lost. If the drive stutters underwater, stop and check sensor lag compensation before increasing current — see Module 4.

> ! To be written by the instructor
>
> The wet-test protocol is course material and has no counterpart in the manufacturing guide. Needed:
>
> - Tank or venue, and supervision requirements
> - Run profile: depth, duration, flap frequency sweep
> - What is logged, and how it is retrieved — SD card handling on a potted board is not trivial, see PCB revisions
> - How wing trim is adjusted, and against what reference
> - Cost of transport measurement procedure, since it is scored in Module 12
> - Recovery and abort procedure

> ℹ Checkpoint — TA sign-off
>
> A completed underwater run with logged data, no water ingress on inspection afterwards, and a recorded cost-of-transport measurement.

13 Module 12 — Competition

Week of Nov 30 · Paired lecture: *Pair presentations*

13.1 What happens today

Robots are scored in two categories: **most efficient** and **fastest**.

13.2 Before you arrive

Bring	Robot, spare crank–linkage shaft, spare wing, charged packs, your logged cost-of-transport data

! To be written by the instructor

Competition format is course material, not in the manufacturing guide. Needed:

- Course layout and run rules for each category
- How cost of transport is computed and normalised across teams with different masses
- Number of attempts, and repair rules between attempts
- How the competition contributes to the grade
- What happens to robots afterwards

13.3 Feedback

The final module is a recap and feedback round. This manual is rebuilt each year from what students report — if something in it cost you time, say so. Corrections made during this semester are listed in the Changelog.

14 Reference — Parts list

Carried over from the manufacturing guide. Quantities are per robot.

Table 4: Parts list

#	Part	Model / material	Mass [g]	Supplier
2	Body plates	CF 1.5 mm		swiss-composites
1	Motor	AT2303 1500 kV		T-Motor · rc-shop.ch
1	Transmission	Faulhaber Series 20/1R 1:23	39	Faulhaber
4	Frame parts	3D-printed Onyx with CF		Markforged
34	Screws	M2×4 countersunk, A4		Bossard A4 or A2
26	Plastic inserts	M2×4 brass		Reichelt
2	Output gears	CNC-cut PEEK		RS Pro
1	Transmission crank	5 mm aluminium		
1	Transmission bearing	4×8 mm flanged stainless	0.6	
1	Linkage	4 mm aluminium		
2	Linkage bearings	3×7×3 mm stainless	0.6	
1	Linkage–gear shaft	3 mm steel	1	
1	Linkage–crank shaft	3 mm steel	0.5	
1	Primary gear shafts	3 mm steel	1.7	
1	Driven gear shafts	3 mm steel	0.8	
1	Battery connector	JWPF 08T & 08R, 8 pins	3.1	Farnell 08R · 08T
1	Voltage regulator	5 V 2 A UBEC	1	Amazon
1	Battery	4S 450 mAh	50.8	Amazon

! To be completed before the semester

Several items the modules rely on are **not** in this list and need adding: the AS2303 motor (the list still shows the older AT2303), the B-G431B-ESC1, the AS5047P sensor, the ST-Link programmer, the FAAV PCB v3.0 and its BOM, the XIAO ESP32S3, 4 mm steel rod for the motor shaft, 5 mm PEEK stock, foam, membrane material, silicone sealant and CorrosionX.

Suppliers are also mostly Swiss, inherited from the EPFL origin of the project. They will need US equivalents for MIT procurement.

15 Reference — Software

15.1 ESC

Code for the B-G431B-ESC1 motor controller using the SimpleFOC library: FAAV-G431-ESC.

Used in Module 3 and Module 4.

15.2 Flight controller

- **Current** — ESP32, based on FreeRTOS tasks: FAAV-Flight-Controller

Outdated versions, kept for reference:

- Old EPFL code for NRF52: FAAV-Seeeduino
- Old MIT code for NRF52/ESP32: FAAV-AUK

15.3 PCB

- FAAV-PCB-v3 — current board, including the BOM in the Manufacturing/Assembly folder

15.4 External documentation

- SimpleFOC communication
- SimpleFOC commander
- XIAO ESP32S3 getting started

> ! Access
>
> Confirm before the semester that students have read access to these repositories, and pin the branch or tag each module expects. “Latest main” changing under a cohort mid-semester is the most likely way this manual goes stale without anyone noticing.

16 Reference — PCB revisions

To interface the MCU to the rest of the FAAV, a custom PCB is required. As of April 2026 the latest board in the FAAV is **v3.0**. More information is in the FAAV-PCB-v3 repository.

Version	Also called	Notes
v1.0	EPFL PCB	Details below
v2.0	MIT PCB	Details below
v3.0	FAAV PCB v3.0	Current board

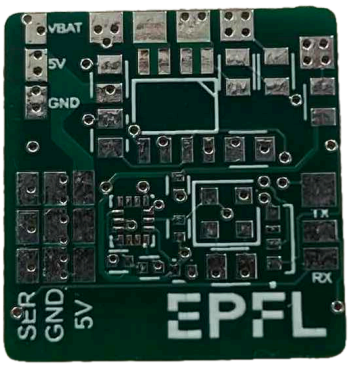


Figure 16: v1.0

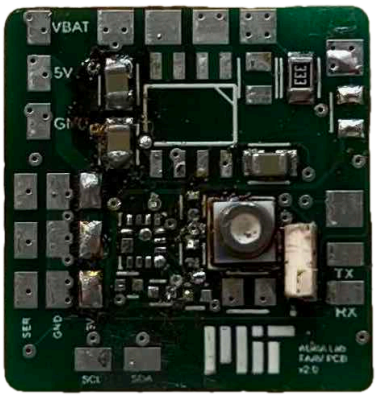


Figure 17: v2.0

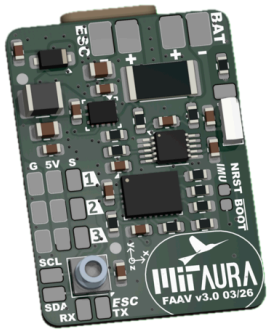


Figure 18: v3.0

16.1 v3.0 overview

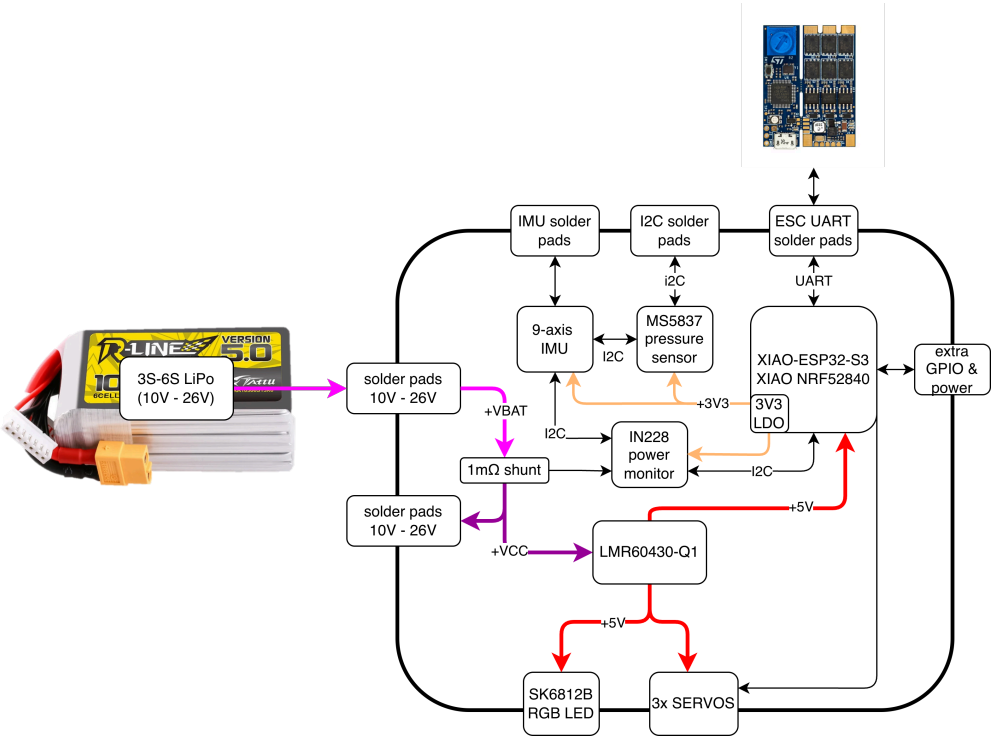


Figure 17: FAAV PCB v3.0 block diagram

The PCB is 4 layers, 21 × 28.5 mm, and has 34 components. The complete BOM is in the Manufacturing/Assembly folder of the repository.

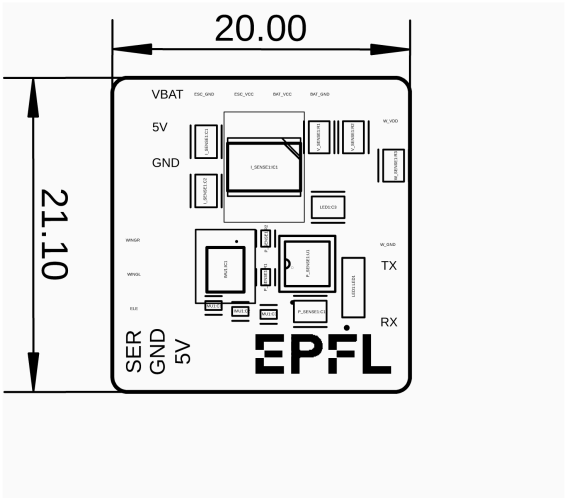
Assembly and bring-up are covered in Module 9.

16.2 v1.0 — EPFL

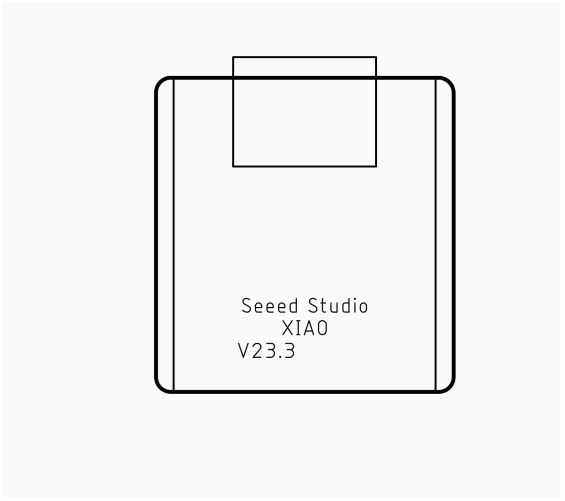
A 2-layer PCB. Its purpose is to connect the flight controller to the battery, ESC and so on, and to provide additional functionality. It was designed with the Fusion360 PCB designer. The PCB contains a current sensor, a voltage sensing circuit, a pressure sensor for measuring water depth, water sensing probes, an LED, an IMU, and several pads for soldering cables. The IMU and current sensor connect to the MCU over I²C.

⚠ Kapton between flight controller and PCB

To prevent short circuits, put a layer of Kapton tape between the flight controller and the PCB when soldering the Seeeduino to the PCB.



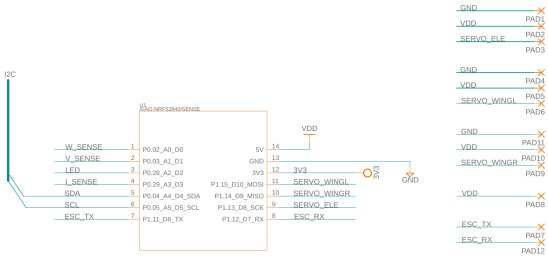
Assembly Drawing Top



Assembly Drawing Bottom

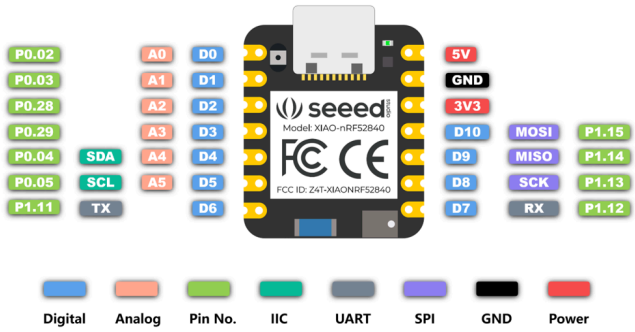
(a) Top (b) Bottom

Figure 18: PCB v1.0 assembly drawings



27.11.23 22:58 /Users/moritzhusser/Desktop/pcb v23.pdf (MCU: Sheet: 1/1)

(a) MCU pin mapping on PCB



(b) Pin map for XIAO nRF52840 (Sense) MCU

Figure 19: Pin mapping of PCB pins and flight controller pins

16.2.a Component list

Table 5: PCB v1.0 components

Names	Component description	Supplier	Item number
IMU1:C1	Capacitor 0.1 µF (0402)	Mouser	581-KGM05AR51A104KH
I_SENSE1:C1, LED1:C3, P_SENSE1:C1	Capacitor 0.1 µF (0805)		
P_SENSE1:R1, P_SENSE1:R2	Resistor 10 k (0402)	Mouser	71-CRCW040210K0FKEDC
IMU1:C2	Capacitor 10 nF (0402)	Mouser	581-KGM05AR71C103KH
I_SENSE1:C2	Capacitor 10 nF (0805)		
W_SENSE1:R3	Resistor 1 M (0805)		
IMU1:C3	Capacitor 2.2 µF (0402)	Mouser	581-KGM05AR51A225KH
V_SENSE1:R1	Resistor 330 k (0805)		
V_SENSE1:R2	Resistor 33 k (0805)		
LED1:LED1	LED 4691 Adafruit	Mouser	485-4691
I_SENSE1:IC1	Current sensor ACS723LLCTR-20AU-T, Allegro	Mouser	250-723LLCTR20AUT
IMU1:IC1	IMU ICM-42688-P, TDK InvenSense	Mouser	410-ICM-42688-P
P_SENSE1:U1	MS583730BA01-50, TE Connectivity	Mouser	824-MS583730BA01-50
I_SENSE1:BAT_GND_REG, I_SENSE1:BAT_VCC_REG	Pin header		
MCU1:U1	XIAO nRF52840 / Sense, Seeed Studio		

16.3 v2.0 — MIT

The goal of this version was to fix existing functional problems (I^2C, pad spacing) and to introduce a new external I^2C pad for GPS functionality. It is also designed to function by default with both the newer ESP32S3 chips and NRF52 chips. Improvements in range through WiFi connection are expected. **Manufacturing steps are identical to v1.0.**

16.3.a ESP32 compatibility

Hardware and software arrangements were needed for XIAO ESP32S3 compatibility. For software see Software.

Hardware changes: a new PCB housing part allowing SD card removal and reinsertion, a WiFi antenna housing case, and a GPS housing case.

SD card notes

1. SD cards should be formatted to a specific format (FAT32, believed).
2. Use non-sharp tweezers to gently extract the SD card from the PCB case — otherwise you risk breaking the card and making it unreadable.
3. For simultaneous servo and SD card usage, be careful not to use the same pins as on the previous chip for servos; TX and RX are used for SD card communication. You may use the secondary microSD card PCB's two additional pins.

16.3.b Waterproofing

The method changes slightly from Module 10. Admittedly it can be vastly improved, but it works:

1. Insert the assembled PCB, making sure the secondary SD card PCB does not come loose, and that the WiFi antenna is properly inserted.
2. Insert the USB-C cable into the port, and a microSD card in the relevant slit.
3. Tape the bottom of the PCB so no silicone escapes when pouring. Pour silicone, using a needle to ensure there are no air bubbles between layers.
4. When fully cured, the SD card is very hard to remove if the protective section broke off during assembly. It can be salvaged by cutting around three sides of the SD card with a knife and opening it like a trap door, freeing the stuck sections. The trap door can then be sealed with epoxy or more silicone — only on the surface, as you do not want to block the SD card again.

The WiFi antenna should also be potted. Currently the same housing case as for the ESC is used: tape the bottom of the box with Kapton tape, stick the antenna to the bottom of the casing (normally all in sticker form), then pot it with silicone. The silicone layer should be left relatively thin.

> ℹ The antenna stops working when wet
>
> It works again once dried. This is expected behaviour, not a fault.

16.3.c Points for a redesign

- Incorrect wiring for pressure sensor — SDA and SCL are swapped.
- Cover the vias (currently open vias).
- Vias on pads of small SMD components are not optimal for soldering, or just cover the vias.
- Increase pad spacing for the cables.
- Current sensor: possibly change to a 3.3 V part. Currently a 5 V sensor is used, so current should not exceed $(3.3\,\mathrm{V} - 0.5\,\mathrm{V})/0.2\,\mathrm{V/A} = 14\,\mathrm{A}$ without risking damage to the analog ports of the Seeeduino.
- ~~UART RX and TX are swapped~~ *(resolved)*

17 Reference — CNC (Stepcraft D660)

Used in Module 5 for the carbon frame, the aluminium crank and linkage, and the PEEK gears.

17.1 Full guide

The complete Stepcraft guide is available as a PDF:

- CNC Stepcraft D600 LIS guide (PDF)
- Original wiki version: wiki.epfl.ch/lis/cnc-stepcraft

> ⚠ Before every job
>
> Make sure the **0 of the machine is at the right location in your file**, and check the orientation of the axes.

17.2 Cutpath generation

Fusion360 generates cutpaths from the parts in the *manufacture* tab, and allows for *manufacturing models*, where parts can be recombined, realigned and even modified. Chamfers on the body plates were added in the manufacturing model rather than in the design.

Operation order: **holes → chamfers → pockets → outline.**

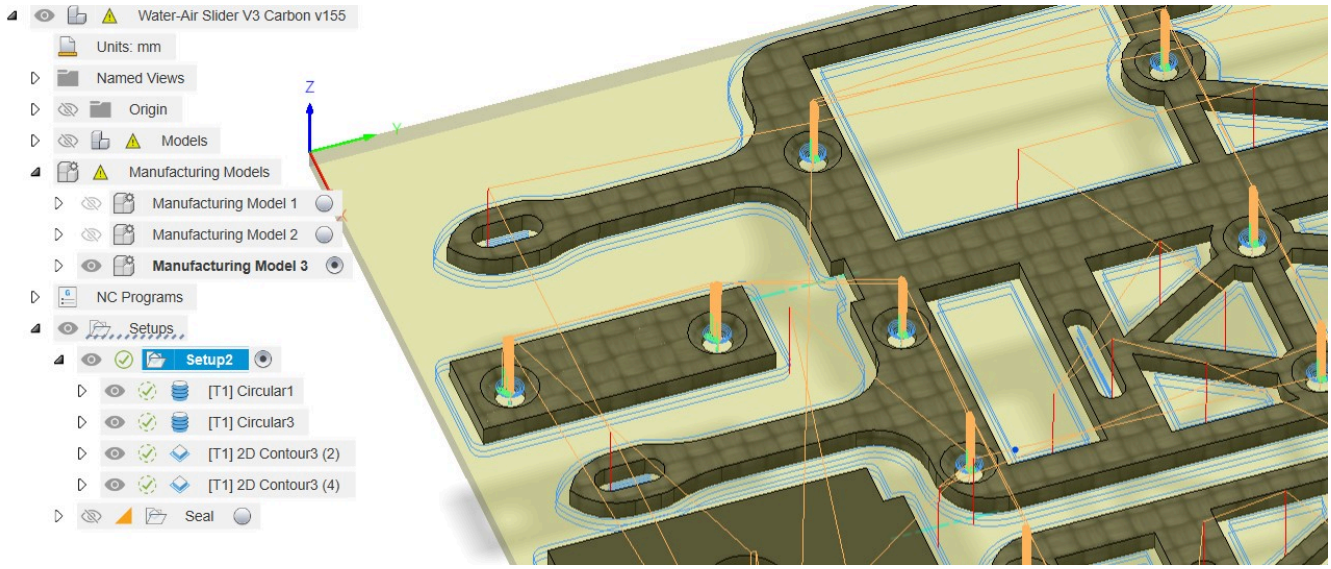


Figure 20: Manufacturing model in Fusion360

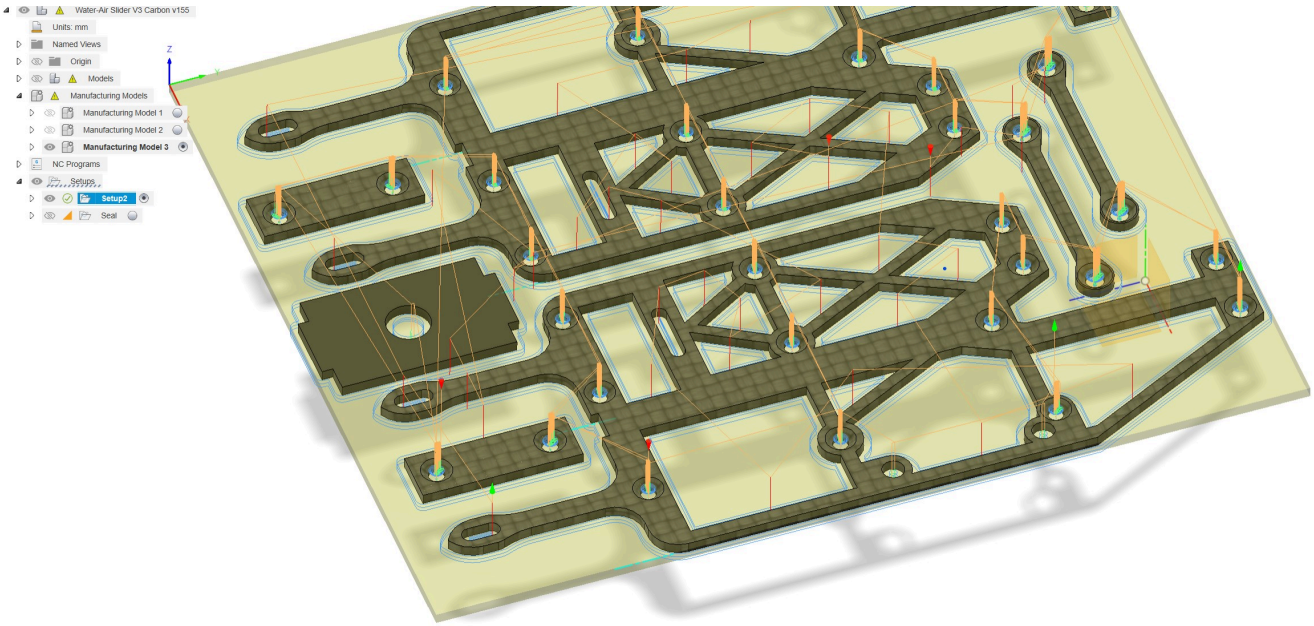


Figure 21: Setting up on the machine

17.3 Material-specific notes

Material	Thickness	Notes
Woven carbon	1.5 mm (varies 1.5–1.6)	Body plates, top and bottom identical except chamfers
Aluminium	5 mm	Crank and linkage. Linkage faced down from 5 to 4 mm. Use a 2 mm endmill, flood the part

Material	Thickness	Notes
PEEK	5 mm	Gears. Cutting underwater gives good results. Keep advance speed low enough to cut teeth cleanly

> ! MIT equivalent
>
> The linked guide is for the EPFL machine. Confirm which machine the class will actually use at MIT and either re-shoot this guide or replace the link.

18 Reference — Design notes

Why the robot is built the way it is. Not needed at the bench — useful for your report, and for anyone changing the design.

18.1 Why an FOC ESC

Several ESC solutions were explored for the FAAV concept:

- **Regular drone ESCs.** Hobbyking 12A and EP Aer11 were tested. Both start **losing synchronisation at low speed**, which is a problem for water propulsion — precisely the regime the robot spends most of its time in.
- **FOC ESCs** tend to be more complex and larger. The **G431-ESC1** board is favourable and had already been explored in the lab.

That tradeoff — accepting a larger, more complex driver to get usable low-speed torque control — is the reason Module 3 spends a whole module on the driver.

18.2 Speed ceiling and sensor lag

Maximum speed went from **500 to 1700 rad/s**, enough to drive the wings at 10 Hz, through a combination of sensor and MCU changes documented in Module 4. The remaining constraint is sensor lag: above 500–600 rad/s, uncompensated lag shows up as high current draw and noise rather than as a hard speed limit.

Compensation is currently hand-tuned every 100 rad/s with a fitted function used on the robot. Automating this — ramping through the speeds while minimising phase current — is open work.

18.2.a Open question

Could `TorqueControlType::foc_current` give better control? The robot currently uses voltage mode, bypassing the current sensors.

18.3 Gearbox corrosion

By default the gearbox on the FAAV4 is not waterproof at all, and it is normal for it to fill completely with water once submerged. This was considered acceptable: over more than a year of repeated use in water, including in Lake Geneva, a gearbox was opened and showed no sign of corrosion.

Then a newer robot, built to experiment with transitions, showed severe corrosion after only **3–4 uses**.



Figure 19: Gearbox after a few dives with motor oil lubrication

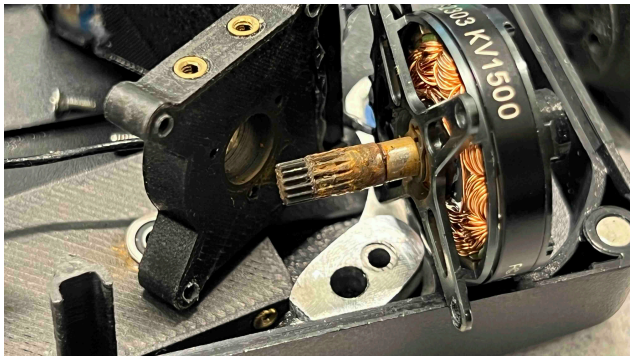


Figure 20: Sun gear from the same gearbox

The cause is believed to be a combination of improper lubrication after a cleaning event, and the absence of the shaft bearing, which allowed more water in through the front of the gearbox. When an old gearbox was cleaned for installation in the new robot, no proper lubricant was available, so motor oil was used. Its low viscosity is believed to have washed it away quickly, leaving the gears dry and starting corrosion.

18.3.a Which parts actually rust

Each individual part was left in water for 12 hours.

Rusts	Does not rust
Sun gear, planet gears, ring gear, planet shafts	Screws, motor shaft, washers, body, circlip

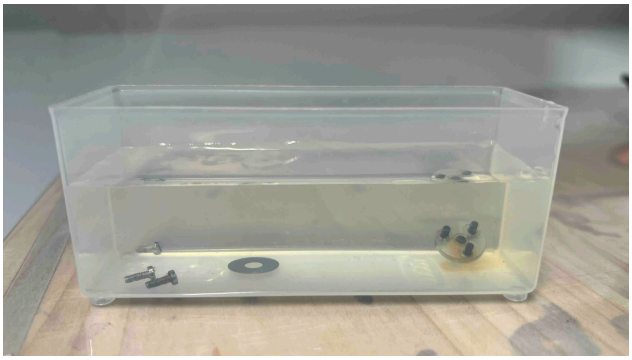


Figure 21: After 12 hours in water

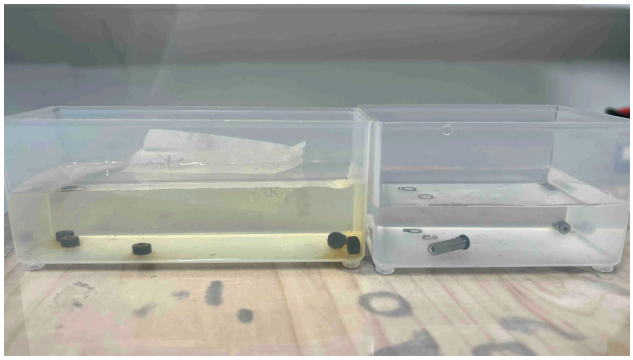


Figure 22: Shafts, sun gear and planet gears show rust

18.3.b Resolution

A few simple optional modifications to the structure and assembly process keep most of the water out. Combined with proper lubrication — a mix of **CorrosionX and white lithium grease** — this appears to have solved the issue. Several dives have since been performed with the same gearbox that had previously corroded, and it now runs smoothly. The practical procedure is in Module 10.

18.4 Housing material tradeoff

Foam	Floatability	Absorption	Impact behaviour
Extruded polystyrene	Best, low density	Zero	Brittle; little help damping landings
Closed-cell polyethylene	Decent, relatively low density	Light	Compliant; usable as protection

A multi-material housing has never been built but would likely be better than either.

19 Changelog

Corrections and additions published during the semester, newest first.

If you are working from a printed copy, check this page before each module. Changes that affect a step you may already have completed are marked in **bold**.

> For maintainers
>
> One row per change: date, page, what changed, and whether teams need to redo anything. Do not edit a step silently — a dated entry is what lets a student at the bench trust the rest of the page.

19.1 fall-2026

Date	Page	Change	Action needed
—	—	Manual published for the Fall 2026 cohort, derived from *FAAV Manufacturing Guide* v4.0	—